

21 JAN 2023

Reg. No. _____

Amrita Vishwa Vidyapeetham
School of Engineering, Bengaluru Campus
B.Tech. Degree Examination - January 2023
Computer Science and Engineering
Fourth Semester

19CSE211 Computer Organization and Architecture

Time: Three hours

Maximum marks: 100

Course Outcomes
Understand the design principles of Instruction Set Architecture (ISA) by taking MIPS as reference.
Design, Implementation and Analysis of data path for instruction execution using Single Clock Cycle
Understand design of instruction execution using Multiple Clock Cycles and Analyze / Evaluate the performance of processors.
Understand Pipelined architecture and Design of 3 and 5 stage pipeline processor in MIPS
Understand the working of Arithmetic and Logic Unit
Understanding the concepts of Memory Organization.

Answer all questions

FOR REFERENCE ONLY

- In MIPS instruction set, top 6 bits (31-26) are used for an opcode. With 6 bits, there are 64 different combinations. Does this imply that total number of instructions in the MIPS architecture is less than 64? [4] [CO1] [BTL4]
- List down the additional registers that are used in multi-cycle data path, over single cycle and also explain how they are used? [4] [CO3] [BTL1]
- Write necessary control signal value for ADDI and J instruction for single cycle MIPS implementation. [4] [CO2] [BTL3]
- What is meant by Average Memory access time in cache memory? How it can be reduced? [4] [CO6] [BTL1]
- Consider a direct mapped cache of size 16 KB with block size of 256 bytes. The size of main memory is 128 KB. Find number of bits required for tag. [4] [CO6] [BTL4]
- How pipelining affect the latency and throughput of a program? [4] [CO4] [BTL1]
- Explain Control Hazard and also Mention different techniques that can be used to deal with control hazard. [4] [CO4] [BTL3]
- Convert 35.75 to its binary representation in IEEE floating point format. [4] [CO5] [BTL5]
- Write the meaning of the following instructions individually independently, given [4] [CO1] [BTL2]
\$s1: 0x00001111, \$s2: 0x11110000, \$s3: 0x01010101
slt \$s1, \$s2, \$s3
nor \$s1, \$s2, \$s3

- 10 If the size of address bus is 35 and data bus is 16, What is the Memory size that can be addressed? What is the size of Program Counter and Instruction Register? Assume each instruction size is 2 bytes. [4]
- 11 (a) Explain the instruction flow of SW instruction using Single cycle data path diagram. [8]
 (b) Write binary format for SW instruction. (opcode SW = 16) [2]
- 12 (a) Assuming a cache has 4096 lines, where each line can store 16 bytes. The main memory has 32-bit address. Find the number of bits in set index and tag for caches that are direct mapped, two-way set associative, four-way set associative, and fully associative. [8]
 (b) What is meant by valid bit and Tag bits in cache memory? [2]
- 13 The clock of the processor runs at 200 MHz. The following table gives instruction frequencies for a Benchmark along with cycles the instructions take, for the different classes of instructions. For this problem, we assume that the processor only executes one instruction at a time. [10]

Instruction Type	Frequency	Cycles
Load & Stores	30%	6 cycle
Arithmetic Instructions	50%	4 cycle
All others	20%	3 cycle

- (a) Calculate the CPI for Benchmark?
- (b) What is the "native MIPS" processor speed for the Benchmark in millions of instructions per second?
- (c) The hardware expert says that if you double the number of registers, the cycle time must be increased by 20%. What would be the new clock speed (in MHz)?
- 14 Consider the following code [10]
- ```

lw $6, 0($4)
lw $4, 5($7)
beq $6, $5, label //assume branch is not taken
sub $7, $6, $2
sw $7, 8($6)

label: sub $6, $7, $6

```
- a. How many clock cycles does it take to execute the code if it is multi cycle execution?
- b. How many clock cycles does it take to execute the code if it is pipelined execution?
- c. In which clock cycle does actual subtraction of \$3, \$2 takes place in case of multi-cycle execution?
- d. In which clock cycle branch target address is calculated in case of multi-cycle execution?
- e. What happens during 8th clock cycle in case of multi-cycle execution?





Amrita Vishwa Vidyapeetham  
Amrita School of Computing, Bengaluru  
B.Tech. Degree Examination - January 2024  
Computer Science and Engineering  
Forth Semester

## 19CSE211 Computer Organization and Architecture

Maximum marks: 100

Time: Three hours

## Course Outcomes

| CO  | Course Outcomes                                                                                                              |
|-----|------------------------------------------------------------------------------------------------------------------------------|
| CO1 | Understand the design principles of Instruction Set Architecture (ISA) by taking MIPS as reference.                          |
| CO2 | Design, Implementation and Analysis of data path for instruction execution using Single Clock Cycle                          |
| CO3 | Understand design of instruction execution using Multiple Clock Cycles and Analyze / Evaluate the performance of processors. |
| CO4 | Understand Pipelined architecture and Design of 3 and 5 stage pipeline processor in MIPS                                     |
| CO5 | Understand the working of Arithmetic and Logic Unit                                                                          |
| CO6 | Understanding the concepts of Memory Organization.                                                                           |

## Answer all questions

- What is meant by pipelining? Explain with the help of an example how pipelining affects the latency and throughput of a processor? [5] [CO4] [BTL2]
- Given a 32 bit processor having total memory size of 2 Terabytes, where each location can store 16 bits of data. What is the size of Address bus and Data bus? [5] [CO6] [BTL4]
- What is the size of Program Counter and Instruction Register?
- Explain any four great ideas in the Computer Architecture? Mention how these ideas are used in practice. [8] [CO1] [BTL3]
- What are the basic components of performance in a given system? [2] [CO1] [BTL3]
- Convert 935.425 to its binary representation in single precision floating point format. Write your answer in hexadecimal notation. [6] [CO5] [BTL3]
- Write the smallest and largest negative number that can be represented in single precision floating point in hexadecimal notation? How NAN and infinity is represented in single precision format? [4] [CO5] [BTL3]
- Write the machine instruction format for the MIPS instruction LW \$2,20(\$2) (opcode for LW=32). Write your answer in hexadecimal. [4] [CO1] [BTL1]
- Computer A has an overall CPI of 1.2 and can be run at a clock rate of 500MHz. Computer B has a CPI of 2.4 and can be run at a clock rate of 750 MHz. We have a particular program to run. When compiled for computer A, this program has exactly 100,000 instructions. How many instructions would the program need to have when compiled for Computer B, in order for the two computers to have exactly the same execution time for this program? [6] [CO1] [BTL3]



- 5 a. Given Direct-mapped cache has 128 blocks/lines, each of size 16 bytes. How the main memory address  $096CFB17_{16}$  is divided into the tag, index & byte offset? [3]  
How does this change if the cache is 4-way set associative?
- b. Consider a Set-associative cache of size 2 KB ( $1 \text{ KB} = 2^{10}$  bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32-bit address is used for accessing the cache. If the width of the tag field is 22 bits, mention the associativity of the cache. [4]
- 6 Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space (RAM) of 64 KB and a word length of 8 bits. During the execution of a program, four data words A, B, C, and D are accessed in that order 3 times (i.e., ABCDABCDABCD). Hence, there are 12 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of A, B, C, and D are  $0 \times A248$ ,  $0 \times C28A$ ,  $0 \times CA8A$ , and  $0 \times A262$ , respectively. Find hit ratio. Tabulate relevant cache entries and justify your conclusion. [10]
- 7 a. Explain the control hazard with an example. Mention any two solutions for dealing with control hazard. [5]
- b. Add comments to the following MIPS code and describe its output in one sentence. Assume the values in \$a0 and \$a1 are the integers a and b, respectively. Assume that \$v0 is used for the output. [5]

```

 ADD $t0, $0, $0
LOOP: BEQ $a1, $0, FINISH
 ADD $t0, $t0, $a0
 SUB $a1, $a1, 1
 J LOOP
FINISH: ADDI $t0, $t0, 100
 ADD $v0, $t0, $0

```

- 8 a. Mention at least two differences among single cycle, multi cycle and pipelined MIPS implementation. [6]
- b. Mention the steps for memory reference instructions in multi cycle MIPS Implementation. [4]
- 9 Explain the instruction flow of beq instruction using single-cycle datapath diagram. Clearly mark size of all the buses and control signals values for the data path. [10]
- 10 The following sequence of instructions are executed on a 5-stage pipelined datapath:
- ```

lw R1,0(R4)
lw R2,400(R4)
add R3,R1,R2
sw R3,0(R4)
addi R4,R4,#1

```

Calculate the number of clock cycles for execution of this segment on the MIPS [5]
 pipeline machine without forwarding or bypassing. Draw the pipeline diagram
 using symbols for the different stages in pipeline
 If forwarding is applied, how many clock cycles are required to execute this code? [5]
 Draw the pipeline data-path to support your answer.

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	25	BTL1	9
CO2	10	BTL2	30
CO3	10	BTL3	36
CO4	20	BTL4	15
CO5	10	BTL5	10
CO6	25	BTL6	0

FOR REFERENCE ONLY

Reg. No.: _____

Anrita Vishwa Vidyapeetham
 School of Computing, Bengaluru Campus
 B.Tech. Degree Examination – December 2024
 Computer Science and Engineering
 Fourth Semester

19CSE211 Computer Organization and Architecture

Time: Three hours

Maximum marks: 100

CO	Course Outcome
CO1	Understand the design principles of Instruction Set Architecture (ISA) by taking MIPS as reference.
CO2	Design, Implementation and Analysis of datapath for instruction execution using Single Clock Cycle.
CO3	Understand design of instruction execution using Multiple Clock Cycles and Analyze / Evaluate the performance of processors.
CO4	Understand Pipelined architecture and Design of 3 and 5 stage pipeline processor in MIPS.
CO5	Understand the working of Arithmetic and Logic Unit.
CO6	Understanding the concepts of Memory Organization.

Answer all questions

- (a) Compare and contrast computer organization and computer architecture. [5] [CO1]
[BTL2]
- (b) Discuss any two factors that affect the performance of a program. [5] [CO1]
[BTL2]
- (a) Which type of instruction set architecture is referred as 1-address type and how [1+3] [CO1]
addition instruction is executed by it? [BTL2]
- (b) Draw the R-Type instruction format and explain each field in it. [3+3] [CO2]
[BTL2]
- (a) Convert the below mentioned MIPS assembly instruction to its binary equivalent [5] [CO2]
machine language. [BTL3]
LW \$t3, 48(\$s4)
- (b) A program runs in 8 seconds on computer X, which has a 3 GHz clock. We are [5] [CO3]
trying to help a computer designer to build a computer, Y, which will run this [BTL3]
program in 4 seconds. The designer has determined that a substantial increase in
the clock rate is possible, but this increase will affect the rest of the CPU design,
causing computer Y to require 1.2 times as many clock cycles as computer X for
this program. What clock rate should we tell the designer to target?

Draw the detailed versatile data-path diagram of single cycle implementation and [10] [CO2]
highlight the path of branch instruction. [BTL4]

5. (a) With the help of a diagram explain how pipelining will increase the overall throughput of a computer system. [3+3]
- (b) How branch prediction will help MIPS pipelining? Discuss. [4]

6. Consider a pipeline having 4 phases with duration 60, 50, 90 and 80 ns. Given latch delay is 10 ns. Calculate the following. (Each sub-question carries two marks) [10]

- Pipeline cycle time
- Non-pipeline execution time
- Speed up ratio
- Pipeline time for 5000 tasks
- Sequential time for 5000 tasks

7. Consider the following code

```
LW $s3, 0($s4)
LW $s4, 5($s7)
BEQ $s3, $s5, Jump //assume branch is not taken
SUB $s7, $s3, $s2
SW $s7, 8($s3)
```

Jump: SUB \$s6, \$s7, \$s3
(Each sub-question carries two marks)

- How many clock cycles does it take to execute the code if it is a multiclock cycle execution?
- How many clock cycles does it take to execute the code if it is a pipelined execution?
- In which clock cycle does actual subtraction of \$3, \$2 takes place in case of multiclock cycle execution?
- In which clock cycle branch target address is calculated in case of multiclock cycle execution?
- What happens during 8th clock cycle in case of multiclock cycle execution?

8. (a) Draw the format of IEEE 754 single precision representation and represent the decimal number 32.75 in IEEE 754 single precision 32-bit format. [2+5] [CO5] [BTL3]

- (b) Discuss how overflow is handled when multiplying two 32-bit numbers. [3] [CO5] [BTL2]

9. (a) How principle of locality will help a computer system to speed-up its memory access process? Explain in detail. [5] [CO6] [BTL2]

(b) Give the definitions of the following memory related keywords.

[5] [CO6]
[BTL1]

- i. Hit rate
- ii. Miss rate
- iii. Hit time
- iv. Miss penalty
- v. Block

10. Consider a direct mapped cache of size 16 KB with a block size of 256 bytes. [7+3] [CO6]
The size of main memory is 128 KB. Find [BTL3]

- i. Number of bits in tag
- ii. Tag directory size.

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	14	BTL1	5
CO2	21	BTL2	42
CO3	5	BTL3	43
CO4	30	BTL4	10
CO5	10	BTL5	0
CO6	20	BTL6	0

Reg. No.: _____

Amrita Vishwa Vidyapeetham
 School of Computing, Bengaluru Campus
 B.Tech. Supplementary Examination – May 2025
 Computer Science and Engineering
 Fourth Semester

19CSE211 Computer Organization and Architecture

Time: Three hours

Maximum marks: 100

CO	Course Outcome
CO1	Understand the design principles of Instruction Set Architecture (ISA) by taking MIPS as reference.
CO2	Design, Implementation and Analysis of datapath for instruction execution using Single Clock Cycle.
CO3	Understand design of instruction execution using Multiple Clock Cycles and Analyze / Evaluate the performance of processors.
CO4	Understand Pipelined architecture and Design of 3 and 5 stage pipeline processor in MIPS.
CO5	Understand the working of Arithmetic and Logic Unit.
CO6	Understanding the concepts of Memory Organization.

Answer all questions

- a) Explain 8 great ideas in computer architecture. [5] [CO1]
[BTL2]
- b) Compare and Contrast CISC and RISC architecture. [5] [CO1]
[BTL2]
- c) Which type of instruction set architecture is referred as 2-address type and how addition instruction is executed by it? [1+3] [CO1]
[BTL2]
- d) Draw the J-Type instruction format and explain each field in it. [3+3] [CO2]
[BTL2]
- e) How data 0A0B0C0D will be stored in memory starting from 0x1000. Express the data in both little endian and big endian. [2.5+ 2.5] [CO2]
[BTL3]
- f) Let us assume that a benchmark has 100 instructions:
 25 instructions are loads/stores (each take 2 cycles)
 50 instructions are add (each takes 1 cycle)
 25 instructions are square root (each takes 50 cycles)
 What is the CPI for this benchmark? [5] [CO3]
[BTL3]
- g) Draw the detailed versatile data-path diagram of single cycle implementation and highlight the path of ADD instruction. [10] [CO2]
[BTL4]

5. (a) Explain various addressing modes available in MIPS with an example for each mode. [6] [CO4] [BTL3]
- (b) Provide any 4 differences between single cycle implementation and multi cycle implementation. [4] [CO4] [BTL2]
6. Consider a non-pipelined processor with a clock frequency of 2.5 gigahertz and average cycles per instruction of 4. The same processor is upgraded to a pipelined processor with five stages but due to the internal pipeline delay, the clock frequency is reduced to 2 gigahertz. Assume there are no stalls in the pipeline and each instruction is executed in 1 clockcycle. The speed up achieved in this pipelined processor is- [10] [CO4] [BTL3]
- Cycle Time in Non-Pipelined Processor
 - Non-Pipeline Execution Time
 - Cycle Time in Pipelined Processor-
 - Pipeline Execution Time
 - Speed up
7. (a) LW R1, 0(R2) [1+4] [CO4]
 ADD R3, R1, R4 [BTL2]
 SUB R5, R3, R6
 AND R1, R5, R7
 OR R8, R1, R9
- Identify what type of hazard is occurring in above example?
 Explain in detail the identified hazard and its pros and cons?
- (b) Explain Structural hazard with example and pipelining diagram. [5] [CO4] [BTL2]
8. (a) Draw the format of IEEE 754 single precision representation and represent the decimal number 28.125 in IEEE 754 single precision 32-bit format. [2+5] [CO5] [BTL3]
- (b) Discuss hierarchy of memory in detail. [3] [CO5] [BTL2]
9. (a) Explain sequential access and hierarchical access with neat diagram. [5] [CO6] [BTL2]
- (b) Differentiate fully associative mapping and set associative mapping with example. [5] [CO6] [BTL2]
10. Consider a fully associative mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find- [7+3] [CO6] [BTL3]
- Number of bits in tag
 - Tag directory size.

Amrita Vishwa Vidyapeetham
Amrita School of Computing, Bengaluru
B.Tech. Degree Examination – June 2023
Computer Science and Engineering
Fourth Semester

19CSE211 Computer Organization and Architecture

Maximum marks: 100

Course Outcomes

- Understand the design principles of Instruction Set Architecture (ISA) by taking MIPS as reference.
Design, Implementation and Analysis of data path for instruction execution using Single Clock Cycle
Understand design of instruction execution using Multiple Clock Cycles and Analyze / Evaluate the performance of processors.
Understand Pipelined architecture and Design of 3 and 5 stage pipeline processor in MIPS
Understand the working of Arithmetic and Logic Unit
Understanding the concepts of Memory Organization.

Answer all questions

- Write the IEEE 754 single precision and double precision floating point format. [4] [CO5]
What is meant by overflow and underflow in single precision notation? [BTL1]
What are the decimal values of the following single precision floats? [4] [CO5]
i. 0x80000000 [BTL3]
ii. 0x41180000
Suppose that instead of 8 bits, exponent is only designated 4 bits. What will be the bias in this case? Why biased bits are used for exponent? [2] [CO5]
[BTL2]
A given processor executes instructions in the following 4 stages (no pipeline), [4] [CO4]
IF, ID, EXE, MEM, WB. Time required for each stage is 42 ns, 25ns, 30ns and [BTL3]
15ns respectively. Compare the performance of a pipelined vs. single cycle implementation of processor if there are 100 instructions in a given sequence.
Mention the sizes of all the buffers used in MIPS Pipelined architecture. List [6] [CO4]
down the contents that will be stored in these buffers to justify your answer. [BTL2]
What is meant by endianness? Explain endianness of MIPS architecture for [5] [CO1]
storing data 0x34762890 starting from address 0x048. [BTL1]
Suppose you wish to run a program P with 6×10^9 instructions on a 5 GHz [5] [CO1]
machine A with a CPI of 0.8, what is the expected CPU time? Calculate the [BTL3]
MIPS ratings for machine A.
What is meant by Pipeline stall? How different types of Hazards avoid the [4] [CO4]
pipeline stalls? [BTL1]

- b. Identify the data hazard in the following instruction sequence.

lw \$1, 100(\$s2)

sub \$5, \$1, \$4

add \$8, \$5, \$3

Find out the number of clock cycles taken for a five stage MIPS pipeline execution with data forwarding in place using time space diagram. Use symbols to represent the stages in pipeline.

- 5 a. Calculate the next instruction address in Instruction memory for the MIPS instruction "beq R0, R1, #0xA048" for following cases. Assume PC: 0x1568A456 and R0:10
- R1=10?
 - R1=5?
- b. Write an assembly program using MIPS instructions which increments i by 2, if j is less than 'k' else 'i' is decremented by 3.
- 6 a. What are the drawbacks of the single cycle implementation? How do we overcome these drawbacks in multicycle implementation?
- b. In a multicycle implementation the time required for the various phases is as: IF=2ns, ID = 2ns, EX=4ns, MEM=4 ns, WB=2ns. Consider the program has 70% ALU instructions, 10% branch instructions, 10% load instructions and 10% store word instructions. Find the execution time for 1000 instructions for this MIPS multicycle design.
- 7 a. A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each. Calculate the total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls.
- b. Caches use the middle range of address bits as the set index and the high order bits as the tag. Why is this done? How might cache performance be affected if the middle bits were used as the tag and the high order bits were used as the set index?
- 8 a. A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. Find the average memory access time in ns.
- b. A certain processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor. In case of directly mapped cache how this will change.

Differentiate among single cycle, multi cycle and pipelined MIPS in terms of organization of various functional units. [6] [CO3]
 Mention the steps for R type instructions and branch instructions in multicycle MIPS Implementation. [4] [CO3]
 [BTL1]

Explain the instruction flow of SW instruction using Single cycle data path diagram. Mention all bus sizes and relevant control signals [8] [CO2]
 [BTL2]

Write binary format for SW instruction. (opcode SW = 16) [2] [CO2]
 [BTL2]

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	20	BTL1	21
CO2	10	BTL2	24
CO3	20	BTL3	45
CO4	25	BTL4	5
CO5	10	BTL5	5
CO6	15	BTL6	0